EAST WEST UNIVERSITY

Lab Report-7

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Section: 01

Submitted To:
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1.Detect the tumor from the images using the segmentation approaches listed below:

(Outline the segmented object to highlight the tumor. You can crop the image for accurate segmentation. i) Similarity approaches:

a) Local/Regional Thresholding, b) Global Thresholding, c) Variable Thresholding, d) Dynamic/Adaptive Thresholding.i) Discontinuity approaches: Edge Detection (Sobel, Canny, Prewitt)

Figure-1

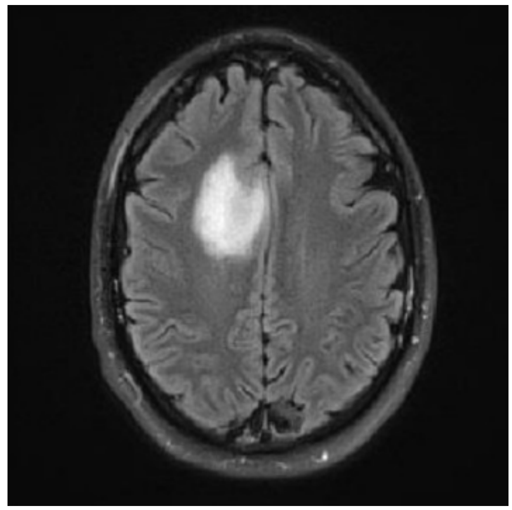


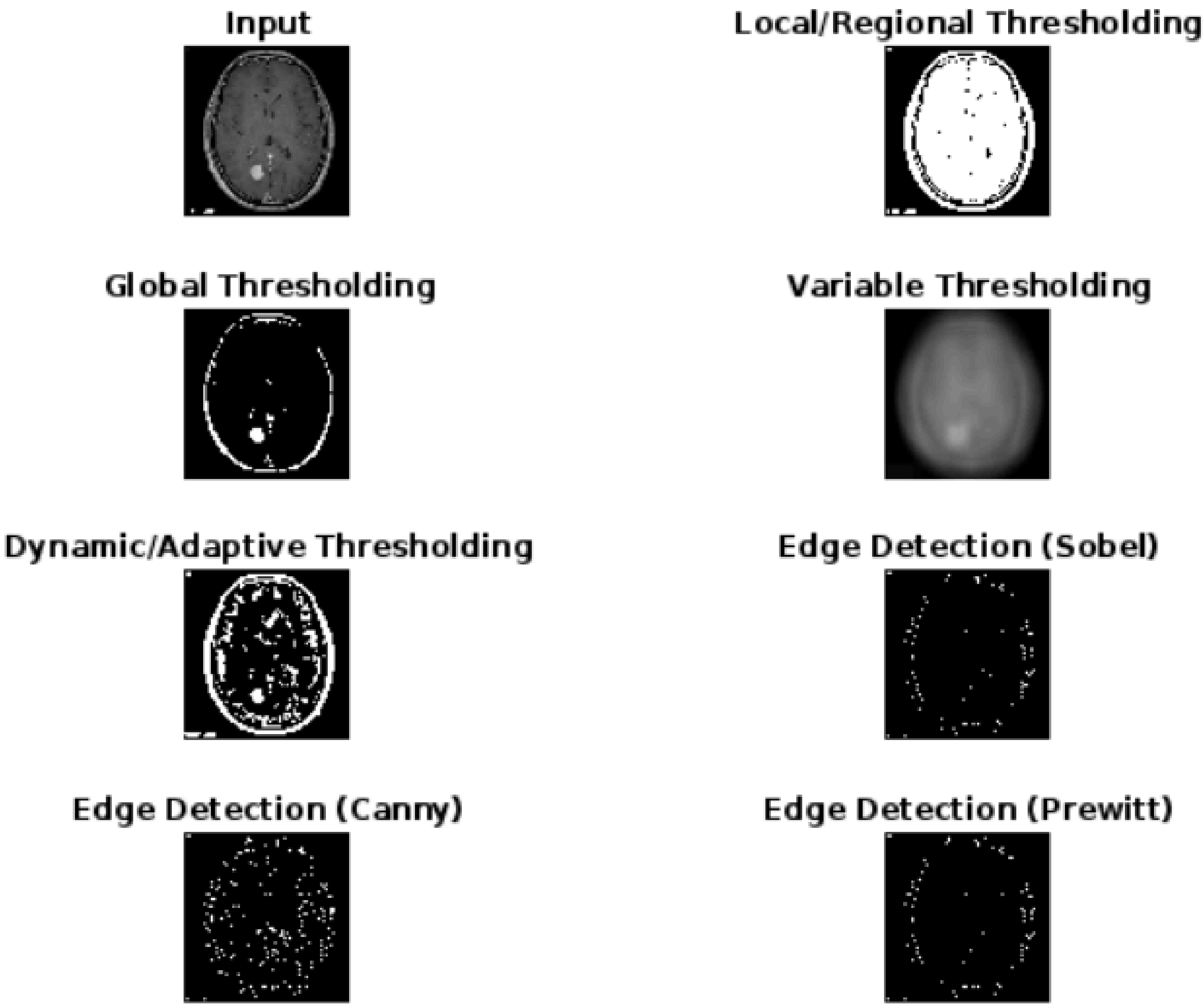
Figure-2

Solution: For figure-1

```
img = imread("picture_7.1.png");

gray_img = rgb2gray(img);

% Apply Gaussian filter

filtered_img = imgaussfilt(gray_img, 3);

% Perform local thresholding

thresh = graythresh(filtered_img);

local_img = imbinarize(filtered_img, thresh);

% Perform global thresholding

thresh =0.4;

global_img = imbinarize(filtered_img, thresh);
```

```matlab
% Perform variable thresholding

var_img = adaptthresh(filtered_img, .2);

% Perform dynamic/adaptive thresholding

adap_img = imbinarize(filtered_img, 'adaptive', 'Sensitivity', 0.5);

% Perform edge detection using the Sobel,Canny,Prewitt operator

sobel_img = edge(filtered_img, 'sobel');

canny_img = edge(filtered_img, 'canny');

prewitt_img = edge(filtered_img, 'prewitt');

%display

subplot(4,2,1),    imshow(img ); title('Input');

subplot(4,2,2),    imshow(local_img); title('Local/Regional Thresholding');

subplot(4,2,3),    imshow(global_img);      title('Global Thresholding');

subplot(4,2,4),    imshow(var_img); title('Variable Thresholding');

subplot(4,2,5),    imshow(adap_img);  title('Dynamic/Adaptive Thresholding');

subplot(4,2,6), imshow(sobel_img ); title('Edge Detection (Sobel)');

subplot(4,2,7), imshow(canny_img ); title('Edge Detection (Canny)');

subplot(4,2,8), imshow(prewitt_img ); title('Edge Detection (Prewitt)');
```

Output:

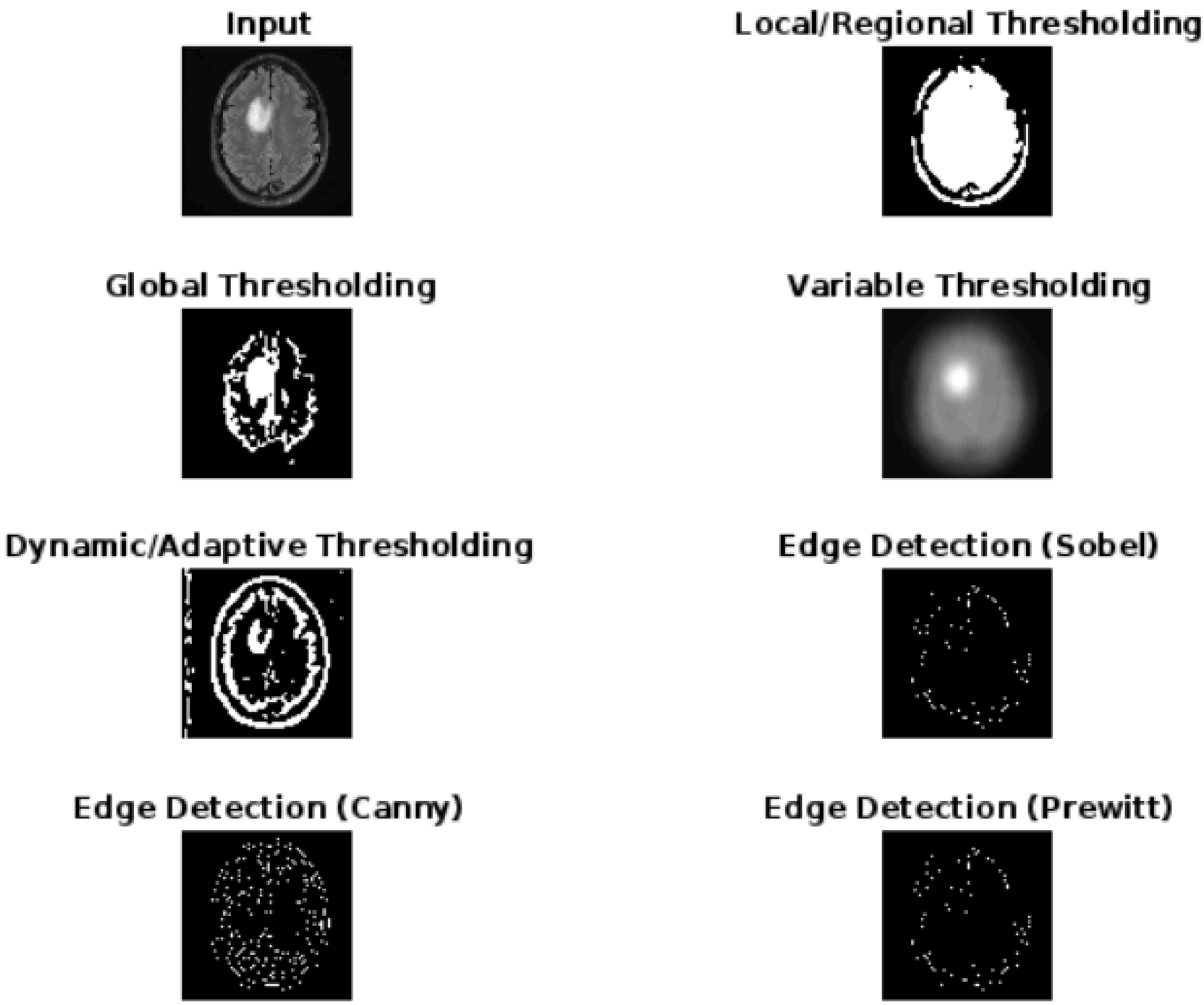


Solution: For figure-2

```
img = imread("picture_7.2.png");

gray_img = rgb2gray(img);

% Apply Gaussian filter

filtered_img = imgaussfilt(gray_img, 3);

% Perform local thresholding

thresh = graythresh(filtered_img);
```

```matlab
local_img = imbinarize(filtered_img, thresh);

% Perform global thresholding

thresh =0.4;

global_img = imbinarize(filtered_img, thresh);

% Perform variable thresholding

var_img = adaptthresh(filtered_img, .2);

% Perform dynamic/adaptive thresholding

adap_img = imbinarize(filtered_img, 'adaptive', 'Sensitivity', 0.5);

% Perform edge detection using the Sobel,Canny,Prewitt operator

sobel_img = edge(filtered_img, 'sobel');

canny_img = edge(filtered_img, 'canny');

prewitt_img = edge(filtered_img, 'prewitt');

%display

subplot(4,2,1),    imshow(img ); title('Input');

subplot(4,2,2),    imshow(local_img); title('Local/Regional Thresholding');

subplot(4,2,3),    imshow(global_img);      title('Global Thresholding');

subplot(4,2,4),    imshow(var_img); title('Variable Thresholding');

subplot(4,2,5),    imshow(adap_img);  title('Dynamic/Adaptive Thresholding');

subplot(4,2,6), imshow(sobel_img ); title('Edge Detection (Sobel)');

subplot(4,2,7), imshow(canny_img ); title('Edge Detection (Canny)');

subplot(4,2,8), imshow(prewitt_img ); title('Edge Detection (Prewitt)');
```

Output:



2. Show in a table how the Similarity and Discontinuity techniques compare.

Solution:

i) For similarity approaches the table is given below,

a) Local/ Regions; Thresholding	b) Global Thresholding	c) Variable Thresholding	d) Dynamic/ Adaptive Thresholding

1.Both the images are brighter.	1. Both the images are darker.	1.Both The images are seen blundering.	1. Both images are seen perfectly.
2. Impossible to identify the tumor.	2. Only the round shaped tumor part is visible.(In figure-1,lower side and in figure-2 upper side)	2. The whole image is blunder and the visible tumor is seen blunder. (In figure-1,lower side and in figure-2 upper side)	2. Appart clearly visible the brain all the edges and round shape tumor((In figure-1,lower side and in figure-2 upper side) which helps the viewer to identify the brain parts.

ii) For discontinuity approaches the table is given below,

a) Sobel edge detection	**b) Canny edge detection**	**c) Prewitt edge detection**
1.Edge detect less compared to canny.	1. Canny detect more edges then sobel and prewitt.	1. Edge detect less compared to canny.
2. Too noise	2. Cleary edges shown in too noise	2.Too noise.
3. Difficult to understand from images.	3. Easy to understand from the pictures.	3. Difficult to understand from images.